Math 1B
Test 3 Study Guide (1.6. 1.7, 2.4, 3.6, 3.7)

Test 3 is **Wednesday 6/17** during class. It covers material from

- **1.6 Integrations involving exponential and logarithms**
- **1.7 Trig Inverse Integration**
- **2.4 Arc Length and Surface Area**
- **3.6 Numerical Integrations**
- **3.7 Improper Integrals**
  - **integration by formulas**
  - **integration by substitution**
  - **integration by parts**
  - **trig integrals**

You can allow my formula pages (derivative/integration formulas and trig unit circle formulas) and one side of your own notes. Scientific and graphing calculators are allowed **except** TI-NSPIRE. Please borrow a calculator from the De Anza reserved desk if you need one. You can't share a calculator, nor use your phone as a calculator.

To better prepare for the test, you should review all lesson **notes**, **Quizzes**, homework **assignments** and end of section exercises from **textbook**.
The following problems give you some examples of problems that you should know. They are NOT test problems.

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1. **Evaluate each indefinite integral.**

$$\int \frac{1}{5^{2x}}\,dx$$

$$\int \frac{\ln x\,dx}{x}$$

$$\int xe^{-x^2}\,dx$$

$$\int \frac{dx}{\sqrt{4-9x^2}}$$

2. **Evaluate each definite integral**

$$\int_{0}^{\pi/4} e^{\cos^2 x} \sin x \cos x \, dx$$

$$\int_{1}^{\sqrt{3}} \frac{1}{\sqrt{4 - x^2}} dx$$

$$\int_{\sqrt{2}}^{2} \frac{1}{|x|\sqrt{x^2 - 1}} dx$$

3. **Evaluate each improper integral or show it's divergent.**

$$\int_{1}^{\infty} \frac{1}{(2x + 1)^3} dx$$

$$\int_{0}^{1} \frac{dx}{\sqrt{1 - x^2}}$$

4. Use **(a)** the Trapezoidal Rule, (**b)** the Mid-point Rule, and **(c)** Simpson's Rule with $n = 6$ to approximate. Round to 6 decimal places.

$$\int_0^{\pi/2} \sqrt{\sin x}\, dx$$

5. a) Find the length of the curve

$$y = \frac{x^3}{6} + \frac{1}{2x} \qquad 1 \leq x \leq 2$$

b) Find the area of the surface obtained by rotating the curve in part (a) about the *x*-axis.

6. Set up integrals of finding the surface area of the volume generated when the following curves revolve around the y-axis

$$y = x + 1 \qquad 1 \leq x \leq 3$$

a) using dx
b) using dy.